LESSON 6.8

Generate Equivalent Expressions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use the commutative, associative, and identity properties to rewrite expressions.

WHY THIS MATTERS You're a band director arranging musicians on stage. You have 3 guitarists, 5 drummers, and 2 keyboard players. Whether you seat the guitarists first or the drummers first, you still have 3 + 5 + 2 = 10 musicians. The order you arrange them doesn't change the total!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Properties of Operations *Spanish: Propiedades de las operaciones*	Rules that let you reorder or regroup numbers without changing an expression's value.	3 + 5 = 5 + 3 (commutative property).
Commutative Property *Spanish: Propiedad conmutativa*	You can change the order and get the same answer.	a + b = b + a
Associative Property *Spanish: Propiedad asociativa*	You can change the grouping and get the same answer.	(a + b) + c = a + (b + c)
Identity Property *Spanish: Propiedad de identidad*	Adding 0 or multiplying by 1 keeps the same value.	a + 0 = a · a × 1 = a
Property *Spanish: Propiedad*	A rule that is always true in math.	Commutative works for + and ×, but NOT for − or ÷ (since 5 − 3 = 2 but 3 − 5 = −2)
Distributive Property *Spanish: Propiedad distributiva*	Multiplying a number by everything inside the parentheses: a(b + c) = ab + ac.	a(b + c) = ab + ac

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Simplify 4(m + 2) + 3m.

1. Distribute the 4: 4m + 8 + 3m.
2. Move like terms together: 4m + 3m + 8.
3. Combine: 7m + 8. Test m = 1: 4(3) + 3 = 15 and 7 + 8 = 15.

Answer: 7m + 8

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Look at 6 + 9 and 9 + 6. Do they give the same total?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I used the _____ Property so that I could _____ .
- After distributing, the expression became _____ .
- The like terms _____ and _____ combine to make _____ .
- My rewritten expression is equivalent because _____ .

Word bank Commutative | Associative | Identity | Distributive | property | equivalent | reorder | regroup | distribute | combine | like terms | value

WATCH OUT!

A common mistake

A common mistake in Properties of Operations is applying the commutative property where it does not belong, or confusing it with the associative property. Students often assume 15 − 8 = 8 − 15 (it does not: 7 ≠ −7

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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Answer: 7m + 8

1 **Circle the letter of the best answer. Show how you know.**

Which property is shown? 5 + 13 = 13 + 5

(A) Commutative Property
(B) Associative Property
(C) Identity Property
(D) Distributive Property

STEPS TO TRY
1. Compare the two sides: same numbers, same operation. What changed between left and right?
2. Match the change to the property: order swapping, regrouping with parentheses, or adding 0 / multiplying by 1.
3. Left is 5 + 13; right is 13 + 5. The addends traded places — find the property about changing order.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which property is shown? (6 × 3) × 2 = 6 × (3 × 2)

(A) Commutative Property
(B) Identity Property
(C) Associative Property
(D) Distributive Property

STEPS TO TRY

1. Read both sides carefully: 6, 3, 2 stay in the same order. Only the parentheses moved.
2. Ask: did the ORDER change, or did the GROUPING change? Each property handles one of those.
3. Left groups (6 × 3) first; right groups (3 × 2) first. Pick the property that's about regrouping.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which property is shown? 47 + 0 = 47

(A) Commutative Property
(B) Associative Property
(C) Distributive Property
(D) Identity Property

STEPS TO TRY

1. Notice what was added: 0. Did adding it change 47 at all?
2. Think about what each property does. Which one is about a number staying itself?
3. 47 + 0 gives back 47 unchanged. Find the property named for keeping a number's identity.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which property lets you rearrange 8 × 5 to 5 × 8?

(A) Associative Property
(B) Identity Property
(C) Distributive Property
(D) Commutative Property

STEPS TO TRY

1. The factors 8 and 5 swapped positions. Nothing else changed — no parentheses, no new numbers.
2. Swapping order is one property; moving parentheses is another. Which happened here?
3. Compare: 8 × 5 became 5 × 8, like trading seats. Choose the property that lets factors change order.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How can you tell the difference between the commutative and associative properties? What clue helps you decide?

SENTENCE STARTER The ___ property changes the ___, while the ___ property changes the ___. I can tell by looking at whether the ___ moved or the ___ changed.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help